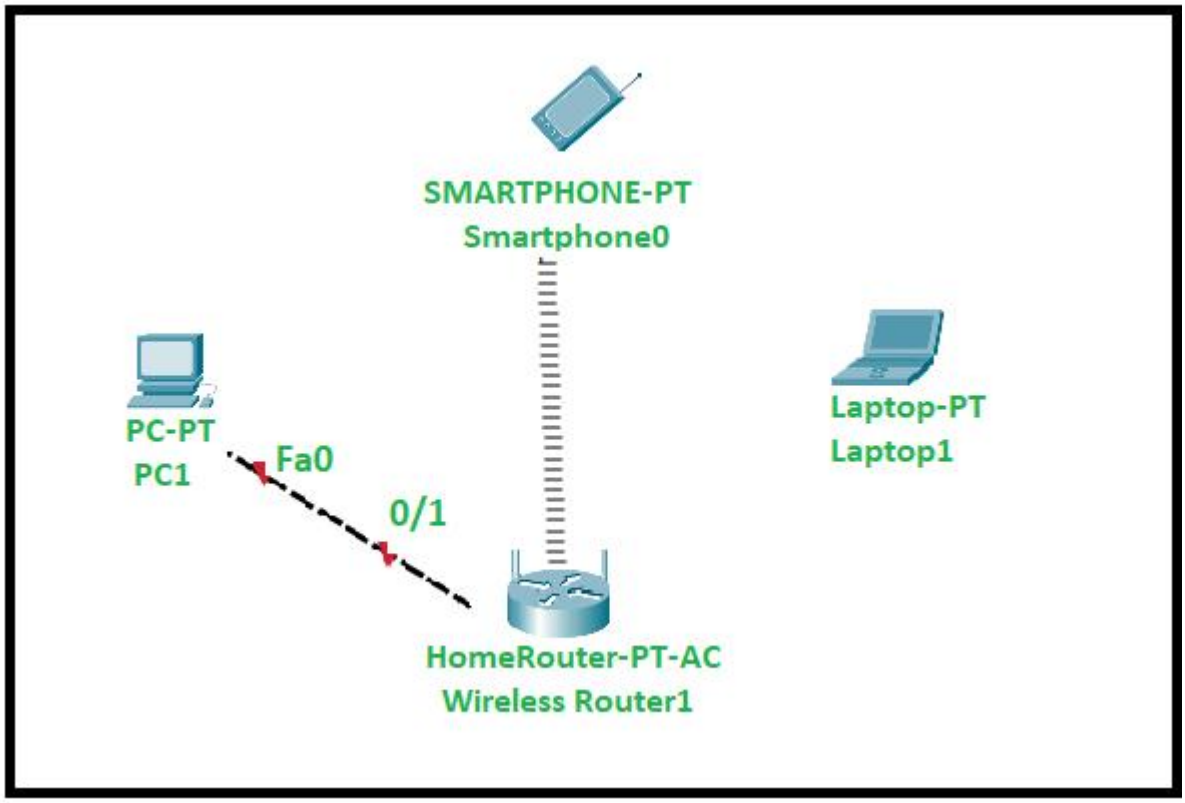
TASK: 1

Open Cisco Packet Tracer and create a new project file named 'HomeWiFiNetwork.pkt' with three PCs and one wireless router connected appropriately.

Aim: To create a simple home network using three PCs and one wireless router in Cisco Packet Tracer.

Network Topology Diagram



Steps:

1. Open Cisco Packet Tracer on your computer.
2. Click **File → New** to create a new project.
3. Add one wireless router and three PCs to the workspace.
4. Rename the devices as PC0, PC1, and PC2.
5. Connect the three PCs to the router using copper straight-through cables and available LAN Ethernet ports. Alternatively, connect them wirelessly if the router and PCs support wireless networking.
6. Click **File → Save As**.
7. Save the project as **HomeWiFiNetwork.pkt**.

Result: A home network containing three PCs and one wireless router is created.

TASK: 2

Assign static IP addresses to each PC in your Packet Tracer network so they are on the same subnet (e.g., 192.168.10.x), and configure the default gateway to the router's IP.

Assign static IP addresses to all three PCs

Use the following IP configuration for your network.

Router LAN IP address: `192.168.10.1`
Subnet mask: `255.255.255.0`

Device	IP Address	Subnet Mask	Default Gateway
PC0	192.168.10.2	255.255.255.0	192.168.10.1
PC1	192.168.10.3	255.255.255.0	192.168.10.1
PC2	192.168.10.4	255.255.255.0	192.168.10.1
Wireless Router (LAN)	192.168.10.1	255.255.255.0	Not applicable

Steps to configure each PC

1. Click **PC0** in Cisco Packet Tracer.
2. Open the **Desktop** tab.
3. Select **IP Configuration**.
4. Select **Static**.
5. Enter:
    - IP Address: `192.168.10.2`
    - Subnet Mask: `255.255.255.0`
    - Default Gateway: `192.168.10.1`
6. Repeat the process for PC1 and PC2, using their respective IP addresses from the table.
7. Open the wireless router's configuration settings and set its LAN IP address to `192.168.10.1` if it is not already configured that way.

Result: All three PCs have unique static IP addresses within the same subnet.

NETWORK

TASK: 3.

Test connectivity by using the Packet Tracer 'ping' tool to verify that all PCs can communicate with each other and the router.

Hint: Use the Command Prompt tool on each PC and type 'ping [other PC's IP]'.

Aim: To verify that all PCs can communicate with one another and with the wireless router.

Ping test from PC0

1. Click PC0.
2. Open **Desktop → Command Prompt**.
3. Type the following commands one at a time:

```
ping 192.168.10.1
ping 192.168.10.3
ping 192.168.10.4
```

Expected output

A successful ping test displays a message similar to:

```
Pinging 192.168.10.3 with 32 bytes of data:

Reply from 192.168.10.3: bytes=32 time<1ms TTL=128
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128

Ping statistics:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)
```

Note: This is sample output. Actual results may differ.

NETWORK

Connectivity testing table

Source Device	Destination	IP Address	Expected Result
PC0	Router	192.168.10.1	Successful
PC0	PC1	192.168.10.3	Successful
PC0	PC2	192.168.10.4	Successful
PC1	PC2	192.168.10.4	Successful

Repeat the ping tests from the other PCs to verify connectivity in both directions.

Result: If the IP addresses, subnet masks, router configuration, and physical or wireless connections are correct, all three PCs should be able to communicate with one another.

TASK: 4.

Document each step you took to build and configure your Packet Tracer network, including screenshots of your topology and the IP configuration windows for each PC, in a Word or PDF file.

NETWORK

Step 1: Create the topology

Open Cisco Packet Tracer, add one wireless router and three PCs, connect the devices, and save the project as `HomeWiFiNetwork.pkt`.

Step 2: Configure PC0

Open PC0 → Desktop → IP Configuration and enter:

- IP Address: `192.168.10.2`
- Subnet Mask: `255.255.255.0`
- Default Gateway: `192.168.10.1`

Step 3: Configure PC1

Open PC1 → Desktop → IP Configuration and enter:

- IP Address: `192.168.10.3`
- Subnet Mask: `255.255.255.0`
- Default Gateway: `192.168.10.1`

Step 4: Configure PC2

Open PC2 → Desktop → IP Configuration and enter:

- IP Address: `192.168.10.4`
- Subnet Mask: `255.255.255.0`
- Default Gateway: `192.168.10.1`

Step 5: Verify connectivity

Open PC0 → Desktop → Command Prompt and run:

```
ping 192.168.10.1
```

```
ping 192.168.10.3
ping 192.168.10.4
```

Step 6: Save the project

Click **File → Save** to save your completed Packet Tracer project.

Final conclusion

A home Wi-Fi network was designed using Cisco Packet Tracer with three PCs and one wireless router. Static IP addresses were assigned to all PCs within the same subnet, and the router was configured as their default gateway. Connectivity was tested using the ping command. Successful ping replies confirmed communication between the PCs and the router.

NETWORK